

2023年11月高数A(I)期中试题解析

一、 填空题(每小题3分, 共18分) .

1. 曲线 $xy - \ln y = 0$ 在点 $(0,1)$ 处的切线方程为 $y = x + 1$.

解析 方程 $xy - \ln y = 0$ 两边对 x 求导得,

易

$$y + xy' - \frac{y'}{y} = 0,$$

将 $x = 0, y = 1$ 代入上式得 $y'(0) = 1$, 故曲线在点 $(0,1)$ 处的切线

方程为 $y - 1 = x$, 即 $y = x + 1$.

2. 函数 $f(x) = \frac{x^2 - x}{|x|(x^2 - 1)}$ 的可去间断点为 1 .

习题册第一章第八节二(2)

解析 函数 $f(x)$ 在 $x = 0, \pm 1$ 处无定义, 且 $x = 0, 1$ 时分子为零.

$$\lim_{x \rightarrow 1} \frac{x(x-1)}{\textcolor{red}{x}(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{2}, \quad x=1 \text{ 为可去间断点;}$$

$$\lim_{x \rightarrow -1} \frac{x(x-1)}{\textcolor{red}{-x}(x-1)(x+1)} = \lim_{x \rightarrow -1} -\frac{1}{x+1} = \infty, \quad x=-1 \text{ 为无穷间断点;}$$

$$\lim_{x \rightarrow 0^-} \frac{x(x-1)}{\textcolor{red}{-x}(x-1)(x+1)} = \lim_{x \rightarrow 0^-} -\frac{1}{x+1} = -1,$$

易

$x=0$ 为跳跃间断点;

$$\lim_{x \rightarrow 0^+} \frac{x(x-1)}{\textcolor{red}{x}(x-1)(x+1)} = \lim_{x \rightarrow 0^+} \frac{1}{x+1} = 1,$$

3. 当 $x \rightarrow 0$ 时, $\sqrt{1+\tan x} - \sqrt{1+\sin x}$ 与 x^n 为同阶无穷小, 则 $n = \underline{\quad 3 \quad}$. 习题册第一章第七节第五题

易

解析

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^n} = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^n (\sqrt{1+\tan x} + \sqrt{1+\sin x})}$$

非零极限因子

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - \sin x}{x^n} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{x^n \cos x}$$

非零极限因子

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x(1 - \cos x)}{x^n} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{x \cdot \frac{1}{2} x^2}{x^n} = \frac{1}{4} \lim_{x \rightarrow 0} x^{3-n} = c \neq 0, \quad 3-n=0, \quad n=3.$$

$$4. \lim_{x \rightarrow 0^+} x \left[\frac{2}{x} \right] = \underline{\hspace{2cm}}.$$

解析 $x \neq 0$ 时, $\frac{2}{x} - 1 < \left[\frac{2}{x} \right] \leq \frac{2}{x}.$

从而当 $x > 0$ 时, $2 - x < x \left[\frac{2}{x} \right] \leq 2$, 由夹逼准则得

$$\lim_{x \rightarrow 0^+} x \left[\frac{2}{x} \right] = 2.$$

5. 设 $f(x) = \lim_{t \rightarrow +\infty} x \left(\frac{t-x}{t+x} \right)^t$, 则 $f^{(n)}(x) = \underline{(-2)^{n-1} e^{-2x} (n-2x)}$

习题册第二章习题课第四题

解析 $f(x) = \lim_{t \rightarrow +\infty} x \left[\left(1 + \frac{-2x}{t+x} \right)^{\frac{t+x}{-2x}} \right]^{\frac{-2xt}{t+x}} = x e^{-2x},$

中

$$\begin{aligned} f^{(n)}(x) &= (x e^{-2x})^{(n)} = (e^{-2x})^{(n)} x + n(e^{-2x})^{(n-1)} \\ &= (-2)^n e^{-2x} \cdot x + n(-2)^{n-1} e^{-2x} = (-2)^{n-1} e^{-2x} (n-2x). \end{aligned}$$

6. 设 $f(x), g(x)$ 互为反函数, 且 $g(1) = 2, g'(1) = 1$, 则

$$d[f(1+x^2)]_{x=1} = \underline{2dx}.$$

解析 $\frac{d}{dx}[f(1+x^2)]_{x=1} = f'(1+x^2) \cdot 2x|_{x=1} = 2f'(2),$

中

由反函数求导得 $f'(2) = \frac{1}{g'(1)} = \frac{1}{1} = 1,$

故 $d[f(1+x^2)]_{x=1} = 2f'(2)dx = 2dx.$

二、判断题(每小题3分, 共15分) .

1. 正数列 $\{x_n\}$ 收敛, 数列 $\{y_n\}$ 发散, 则数列 $\{x_n \cdot y_n\}$ 一定发散. $\times$

解析 例如取数列 $\{x_n\} = \left\{ \frac{1}{n} \right\}$, $\{y_n\} = \{\sin n\}$,

显然 $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$, 数列 $\{y_n\} = \{\sin n\}$ 发散, 但

$$\lim_{n \rightarrow \infty} x_n \cdot y_n = \lim_{n \rightarrow \infty} \frac{1}{n} \sin n = 0.$$

2. 设 $f(x)$ 在点 x_0 可导, 则 $\exists \delta > 0$, 使 $f(x)$ 在 $(x_0 - \delta, x_0 + \delta)$ 连续. $\times$

解析 $f(x) = x^2 D(x)$, $D(x) = \begin{cases} 1, & x \in \mathbf{Q}, \\ 0, & x \in \mathbf{R} / \mathbf{Q}. \end{cases}$

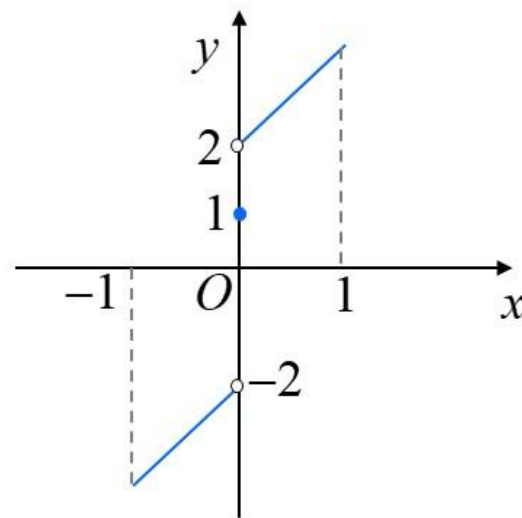
$$\lim_{x \rightarrow 0} \frac{x^2 D(x) - 0}{x} = \lim_{x \rightarrow 0} x D(x) = 0 = f'(0).$$

但 $f(x)$ 在其他点均不连续.

3. 设 $f(x)$ 在 $[a, b]$ 单调有界, 且 $f(a) \cdot f(b) < 0$, 则存在 $\xi \in (a, b)$, 使得 $f(\xi) = 0$. $\times$

解析 例

$$f(x) = \begin{cases} x+2, & 0 < x \leq 1, \\ 1, & x = 0, \\ x-2, & -1 \leq x < 0. \end{cases}$$



显然 $f(x)$ 在 $[-1, 1]$ 单调有界, 且 $f(-1) \cdot f(1) < 0$, 但不存在 $\xi \in (-1, 1)$, 使得 $f(\xi) = 0$.

4. 若 $f(x)$ 在 $[a, +\infty)$ 上无界, 则对 $\forall G > 0, \exists X > 0$, 当 $x > X$ 时, 有 $|f(x)| > G$. ✕

解析 $f(x)$ 在 $[a, +\infty)$ 上无界 $\iff \forall G > 0$, 总 $\exists x_0 \in [a, +\infty)$, 使得 $|f(x)| > G$.

$\lim_{x \rightarrow +\infty} f(x) = \infty \iff \forall G > 0, \exists X > 0$, 当 $x > X$ 时, $|f(x)| > G$.

5. 设 $f(x)$ 连续, 且 $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h} = 2023$, 则 $f'(a) = 2023$. ❌

解析 例 $f(x) = \begin{cases} 4046x, & x > 0, \\ 0, & x \leq 0. \end{cases}$ 在 $x=0$ 连续但不可导, 而

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0-h)}{2h} = \lim_{h \rightarrow 0^-} \frac{f(h) - f(-h)}{2h} = \lim_{h \rightarrow 0^-} \frac{0 - 4046(-h)}{2h} = 2023,$$

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0-h)}{2h} = \lim_{h \rightarrow 0^+} \frac{f(h) - f(-h)}{2h} = \lim_{h \rightarrow 0^+} \frac{4046h - 0}{2h} = 2023,$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h} = 2023.$$

三、 计算题(每小题7分, 共28分) .

1. 求 $\lim_{n \rightarrow \infty} \left(\frac{1}{n(\sqrt{n^2+1}+n)} + \frac{2}{n(\sqrt{n^2+2}+n)} + \cdots + \frac{n}{n(\sqrt{n^2+n}+n)} \right).$

解析 令 $a_n = \frac{1}{n(\sqrt{n^2+1}+n)} + \frac{2}{n(\sqrt{n^2+2}+n)} + \cdots + \frac{n}{n(\sqrt{n^2+n}+n)}$

$$= \frac{1}{n^2 \left(\sqrt{1 + \frac{1}{n^2}} + 1 \right)} + \frac{2}{n^2 \left(\sqrt{1 + \frac{2}{n^2}} + 1 \right)} + \cdots + \frac{n}{n^2 \left(\sqrt{1 + \frac{n}{n^2}} + 1 \right)},$$

$$b_n = \frac{1}{n^2 \left(\sqrt{1 + \frac{1}{n^2}} + 1 \right)} + \frac{2}{n^2 \left(\sqrt{1 + \frac{1}{n^2}} + 1 \right)} + \cdots + \frac{n}{n^2 \left(\sqrt{1 + \frac{1}{n^2}} + 1 \right)},$$

$$c_n = \frac{1}{n^2 \left(\sqrt{1 + \frac{n}{n^2}} + 1 \right)} + \frac{2}{n^2 \left(\sqrt{1 + \frac{n}{n^2}} + 1 \right)} + \cdots + \frac{n}{n^2 \left(\sqrt{1 + \frac{n}{n^2}} + 1 \right)},$$

显然 $c_n \leq a_n \leq b_n$.

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1+2+\cdots+n}{n^2 \left(\sqrt{1+\frac{1}{n^2}} + 1 \right)} = \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2 \left(\sqrt{1+\frac{1}{n^2}} + 1 \right)} = \frac{1}{4},$$

$$\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \frac{1+2+\cdots+n}{n^2 \left(\sqrt{1+\frac{n}{n^2}} + 1 \right)} = \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2 \left(\sqrt{1+\frac{n}{n^2}} + 1 \right)} = \frac{1}{4},$$

由夹逼准则, $\lim_{n \rightarrow \infty} a_n = \frac{1}{4}$.

2. 求 $\lim_{x \rightarrow 0^+} \frac{4}{x} \left(e^x - \sqrt{\frac{\sin x}{\ln(1+x)}} \right).$

解析 $\lim_{x \rightarrow 0^+} \frac{4}{x} \left(e^x - \sqrt{\frac{\sin x}{\ln(1+x)}} \right) = \lim_{x \rightarrow 0^+} \frac{4}{x} \left[(e^x - 1) - \left(\sqrt{\frac{\sin x}{\ln(1+x)}} - 1 \right) \right]$

$$= \lim_{x \rightarrow 0^+} 4 \left[\frac{e^x - 1}{x} - \frac{1}{x} \left(\sqrt{\frac{\sin x}{\ln(1+x)}} - 1 \right) \right],$$

而 $\lim_{x \rightarrow 0^+} 4 \frac{e^x - 1}{x} = 4,$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{4}{x} \left(\sqrt{\frac{\sin x}{\ln(1+x)}} - 1 \right) &= \frac{4}{x} \lim_{x \rightarrow 0^+} \frac{\frac{\sin x}{\ln(1+x)} - 1}{\sqrt{\frac{\sin x}{\ln(1+x)}} + 1} = 2 \lim_{x \rightarrow 0^+} \frac{\sin x - \ln(1+x)}{x \ln(1+x)} \\ &= 2 \lim_{x \rightarrow 0^+} \frac{\sin x - \ln(1+x)}{x^2} = 2 \lim_{x \rightarrow 0^+} \frac{x + o_1(x^2) - \left[x - \frac{x^2}{2} + o_2(x^2) \right]}{x^2} = 1, \end{aligned}$$

$$\text{故 } \lim_{x \rightarrow 0^+} \frac{4}{x} \left(e^x - \sqrt{\frac{\sin x}{\ln(1+x)}} \right) = 4 - 1 = 3.$$

3. 求 $\lim_{n \rightarrow \infty} n^2 \left(\sin \frac{\pi}{n} - \sin \frac{\pi}{n+1} \right)$.

解析 法一 微分中值定理

$$\lim_{n \rightarrow \infty} n^2 \left(\sin \frac{\pi}{n} - \sin \frac{\pi}{n+1} \right) = \lim_{n \rightarrow \infty} n^2 \cdot \cos \xi \cdot \left(\frac{\pi}{n} - \frac{\pi}{n+1} \right)$$

$$\left(\frac{\pi}{n+1} < \xi < \frac{\pi}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \cos \xi \cdot n^2 \frac{(n+1)\pi - n\pi}{n(n+1)} = \lim_{\substack{n \rightarrow \infty \\ \xi \rightarrow 0}} \cos \xi \cdot \frac{n^2 \pi}{n(n+1)} = 1 \times \pi = \pi.$$

法二 和差化积公式

$$\begin{aligned}\lim_{n \rightarrow \infty} n^2 \left(\sin \frac{\pi}{n} - \sin \frac{\pi}{n+1} \right) &= \lim_{n \rightarrow \infty} n^2 \cdot 2 \cos \frac{\frac{\pi}{n} + \frac{\pi}{n+1}}{2} \sin \frac{\frac{\pi}{n} - \frac{\pi}{n+1}}{2} \\&= \lim_{n \rightarrow \infty} 2n^2 \cdot \cos \frac{\frac{\pi}{n} + \frac{\pi}{n+1}}{2} \cdot \frac{\pi}{2n(n+1)} \\&= \lim_{n \rightarrow \infty} \cos \frac{\frac{\pi}{n} + \frac{\pi}{n+1}}{2} \cdot \frac{2n^2 \pi}{2n(n+1)} = \cos 0 \times \pi = \pi.\end{aligned}$$

4. 求 $\lim_{x \rightarrow 0} \left[\frac{\tan x}{\ln(1+x)} \right]^{\frac{1}{x}}$.

解析 令 $y = \lim_{x \rightarrow 0} \left[\frac{\tan x}{\ln(1+x)} \right]^{\frac{1}{x}}$,

$$\ln y = \frac{1}{x} \ln \frac{\tan x}{\ln(1+x)} = \frac{1}{x} \ln \left[1 + \frac{\tan x - \ln(1+x)}{\ln(1+x)} \right],$$

$$\begin{aligned} \lim_{x \rightarrow 0} \ln y &= \lim_{x \rightarrow 0} \frac{1}{x} \ln \left[1 + \frac{\tan x - \ln(1+x)}{\ln(1+x)} \right] = \lim_{x \rightarrow 0} \frac{1}{x} \cdot \frac{\tan x - \ln(1+x)}{\ln(1+x)} \\ &= \lim_{x \rightarrow 0} \frac{\tan x - \ln(1+x)}{x^2} \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x - \ln(1+x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x - \frac{1}{1+x}}{2x} = \lim_{x \rightarrow 0} \frac{2\sec^2 x \tan x + \frac{1}{(1+x)^2}}{2} = \frac{1}{2},$$

故 $\lim_{x \rightarrow 0} \left[\frac{\tan x}{\ln(1+x)} \right]^{\frac{1}{x}} = e^{\frac{1}{2}}.$

解析

$$\begin{aligned}\lim_{x \rightarrow 0} \left[\frac{\tan x}{\ln(1+x)} \right]^{\frac{1}{x}} &= \lim_{x \rightarrow 0} \left\{ \left[1 + \frac{\tan x - \ln(1+x)}{\ln(1+x)} \right]^{\frac{\ln(1+x)}{\tan x - \ln(1+x)}} \right\}^{\frac{\tan x - \ln(1+x)}{x \ln(1+x)}} \\ &= \lim_{x \rightarrow 0} e^{\frac{\tan x - \ln(1+x)}{x \ln(1+x)}} = \lim_{x \rightarrow 0} e^{\frac{[x + o_1(x^2)] - [x - \frac{1}{2}x^2 + o_2(x^2)]}{x^2}} = e^{\frac{1}{2}}.\end{aligned}$$

四、解答题(每小题7分, 共21分) .

1. 设 $\lim_{x \rightarrow \infty} \left[x \ln \left(e + \frac{1}{x} \right) - ax - b \right] = 0$, 求常数 a 和 b 的值.

解析 $\lim_{x \rightarrow \infty} \left[x \ln \left(e + \frac{1}{x} \right) - ax - b \right] = \lim_{x \rightarrow \infty} x \left[\ln \left(e + \frac{1}{x} \right) - x - \frac{b}{x} \right] = 0,$

$$\Rightarrow \lim_{x \rightarrow \infty} \left[\ln \left(e + \frac{1}{x} \right) - a - \frac{b}{x} \right] = 0, \quad \Rightarrow \quad 1 - a = 0, \quad a = 1.$$

$$\Rightarrow b = \lim_{x \rightarrow \infty} \left[x \ln \left(e + \frac{1}{x} \right) - x \right] = \lim_{x \rightarrow \infty} x \left[\ln \left(e + \frac{1}{x} \right) - \ln e \right]$$

$$b = \lim_{x \rightarrow \infty} \left[x \ln \left(e + \frac{1}{x} \right) - x \right] = \lim_{x \rightarrow \infty} x \left[\ln \left(e + \frac{1}{x} \right) - \ln e \right]$$

$$= \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{1}{ex} \right) \lim_{x \rightarrow \infty} x \frac{1}{ex} = \frac{1}{e}.$$

2. 设 $\begin{cases} x = 3t^2 + 2t, \\ e^y \sin t - y + 1 = 0, \end{cases}$ 求 $\left. \frac{dy}{dx} \right|_{t=0}$.

解析 $\frac{dx}{dt} = 6t + 2, \quad e^y \frac{dy}{dt} \sin t + e^y \cos t - \frac{dy}{dt} = 0,$

$t = 0 \Rightarrow x = 0, y = 1,$ 从而 $\left. \frac{dx}{dt} \right|_{t=0} = 2, \quad \left. \frac{dy}{dt} \right|_{t=0} = e,$

所以 $\left. \frac{dy}{dx} \right|_{t=0} = \frac{\left. \frac{dy}{dt} \right|_{t=0}}{\left. \frac{dx}{dt} \right|_{t=0}} = \frac{e}{2}.$

3. 设对任意实数 x, y 都有 $f(x+y) = f(x)f(y)$, 且 $f'(0) = 1$, 证明: $f(x)$ 在 $(-\infty, +\infty)$ 上可导, 并求 $f(x)$.

解析 $f(0) = f(0+0) = [f(0)]^2$, $f(0) = 0$ 或 $f(0) = 1$.

若 $f(0) = 0$, 则 $f(x) = f(x+0) = f(x)f(0) = 0$, $f'(0) = 0$,

与 $f'(0) = 1$ 矛盾, 故 $f(0) = 1$.

$$\begin{aligned} \text{任取 } x \in (-\infty, +\infty), \quad f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} f(x) \frac{f(\Delta x) - f(0)}{\Delta x} = f(x) \cdot f'(0) = f(x), \end{aligned}$$

$$f(x) - f'(x) = 0,$$

$$\Rightarrow \frac{d}{dx} \left(\frac{f(x)}{e^x} \right) = \frac{f'(x)e^x - f(x)e^x}{e^{2x}} = 0,$$

$$\Rightarrow \frac{f(x)}{e^x} = C, \quad f(x) = C e^x,$$

$$f'(0) = 1, \Rightarrow f(x) = e^x.$$

五、(12分) (1) 设 $x > 0$, 证明: $x < e^x - 1 < xe^x$.

(2) 设 $x_1 > 0$, $x_n e^{x_{n+1}} = e^{x_n} - 1, n = 1, 2, \dots$, 证明: 数列 $\{x_n\}$ 收敛,
并求 $\lim_{n \rightarrow \infty} x_n$.

解析 (1) $e^x - 1 = e^x - e^0 = xe^\xi, \xi \in (0, x) \Rightarrow e^0 < e^\xi < e^x$

$$\Rightarrow 1 < e^\xi < e^x \Rightarrow x < xe^\xi < xe^x \Rightarrow x < e^x - 1 < xe^x, \quad x > 0.$$

(2) $x_1 > 0, e^{x_2} = \frac{e^{x_1} - 1}{x_1} > 1 \Rightarrow x_2 > 0$, 假设 $x_k > 0$, 则由(1)知

$$e^{x_{k+1}} = \frac{e^{x_k} - 1}{x_k} > 1 \Rightarrow x_{k+1} > 0, \text{由数学归纳法知 } x_n > 0, n \in \mathbf{Z}^+.$$

再由(1)知

$$x_n e^{x_{n+1}} = e^{x_n} - 1 < x_n e^{x_n} \Rightarrow e^{x_{n+1}} < e^{x_n} \Rightarrow x_{n+1} < x_n,$$

从而数列 $\{x_n\}$ 单调递减. 由单调有界准则, $\{x_n\}$ 收敛.

设 $\lim_{n \rightarrow \infty} x_n = a$, 对 $x_n e^{x_{n+1}} = e^{x_n} - 1$ 两边取极限, 得

$$ae^a = e^a - 1,$$

解得 $a = 0$, 故 $\lim_{n \rightarrow \infty} x_n = 0$.

六、(6分) 设 $f(x)$ 在 $[0,1]$ 上连续, 在 $(0,1)$ 内可导, 且 $f(0)=0$, $f(1)=1$, 证明 (1) 存在 $c \in (0,1)$, 使得 $f(c)=1-c$;

(2) 存在 $\xi, \eta \in (0,1)$, 且 $\xi \neq \eta$, 使得 $f'(\xi) \cdot f'(\eta) = 1$.

解析 (1) 令 $F(x) = f(x) + x - 1$, 则 $F(x) \in C[0,1]$, $F(0) \cdot F(1) = -1 < 0$, 由零点定理可知, 存在 $c \in (0,1)$, 使得 $f(c) = 1 - c$;

(2) 由拉格朗日中值定理得, $\exists \xi \in (0,c), \exists \eta \in (c,1)$, 使得

$$f'(\xi) = \frac{f(c) - f(0)}{c} = \frac{1-c}{c}, \quad f'(\eta) = \frac{f(1) - f(c)}{1-c} = \frac{c}{1-c},$$

$$\Rightarrow f'(\xi) \cdot f'(\eta) = 1.$$